

Ali Zain

AI & Machine Learning Engineer

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Professional Summary

AI/ML Engineer with 2+ years of experience developing and deploying end-to-end LLM applications, ML model training, RAG pipelines, and computer vision systems. Proficient in FastAPI microservices, vector search indexing, and AWS cloud deployment. Track record of cutting manual workflows by 60–70% across legal and fintech domains through structured data extraction, tool-calling agents, and custom object-detection models.

Work Experience

Machine Learning & Python Developer

Intalytic Group Ltd, Islamabad

November 2024 – March 2026

- Trained and optimized a YOLOv8 person-detection model for squash-court analytics (annotation to evaluation) and a Random Forest speech-confidence classifier at 84% accuracy.
- Built document processing pipeline and led API development for OCR, extraction, validation, and automated storage of banking documents to AWS S3, reducing manual verification workload by 60%.
- Developed an AI-powered timesheet-to-payroll pipeline for [payseye.com](#), migrating from GPT-4o to GPT-5.4-mini for structured data extraction and optimizing via few-shot prompting and chunking to reduce hallucination; deployed to AWS with a 4+ member team.

AI Developer (Contract)

Ikram Associates, Rawalpindi

February 2024 – October 2024

- Built SaaS platform for law firms to automate document workflows reducing case processing time by 40%.
- Optimized Small Language Model (SLM) inferences using few-shot prompting and structured context formatting to tailor outputs for legal document generation.

Projects

PPE Detection for Construction Safety | YOLOv8, PyTorch, Ultralytics, OpenCV

- Fine-tuned YOLOv8 object detector across 10 PPE classes achieving 0.81 mAP@0.5 and 90% precision.

AI SEO-Blog Generation | LangGraph, Anthropic, OpenAI, Python, FastAPI

- SEO-ready AI blog generation workflow using LLMs (Anthropic, OpenAI) cutting manual writing effort by ~80%.

B2B and B2C Chatbots | Python, OpenAI, FastAPI, MongoDB

- Architected tool-calling agents via OpenAI API with MongoDB aggregation pipelines, supporting concurrent users.

RAG Fashion Bot | LangChain, Weaviate, OpenAI

- Built RAG chatbot for fashion knowledge-base search across 44,000+ products, handling real-time customer Q&A.

EDUCATION

Bachelor of Science in Information Technology

Quaid-i-Azam University, Islamabad – CGPA 3.3/4.0

Graduated: 2025

SKILLS & Certifications

AI/ML: Scikit-learn, TensorFlow, PyTorch, LangChain, LangGraph, RAG, YOLO, OpenAI, Hugging Face

Backend & Tools: Python, SQL, FastAPI, Docker, AWS, PostgreSQL, MongoDB, Pgvector, Weaviate, FAISS, Git

[Machine Learning Specialization](#) | [Retrieval Augmented Generation](#) | [Microsoft Certified: Azure AI Fundamentals](#)